

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limit:

(a) $\lim_{x \rightarrow +\infty} \sqrt{\frac{25x^2 - 4}{4x^2 + 81}}.$ [4 marks]

(b) $\lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}.$ [4 marks]

2. By using implicit differentiation for the function $3x^2 + y^2 = 2$, show that $\frac{d^2y}{dx^2} = -\frac{6}{y^3}.$ [9 marks]

3. If $y = (x^2 - 4)^{\frac{1}{3}}$, find the intervals where the function is increasing and decreasing. [8 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. Given $\bar{z}$ is the conjugate of a complex number z such that $\bar{z} = \sqrt{21 + 20i}$. Find the complex number z such that its real part is negative. [5 marks]
2. (a) Solve the equation $x^6 e^{-4 \ln x} = 2x + 15$. [5 marks]
(b) By completing the square, express the inequality $x^2 + 10x + r \leq 0$ in the form $|x + h| \leq k$, where h , k and r are constants. Hence, find the set of values of x such that $x^2 + 10x + 9 \leq 0$. [7 marks]
3. Given that $f(x) = \frac{2x}{x-1}$.
(a) Determine the composite function $f \circ f$. State the values of x such that $f \circ f$ is undefined. [3 marks]
(b) Find the inverse function, $f^{-1}(x)$ and state its domain. [4 marks]
(c) Find $g(x)$ such that $g \circ f(x) = \frac{8x^2 - 4x}{(x-1)^2}$. [6 marks]
4. Function $g(x)$ is defined by $g(x) = 1 + 2^{x-2}$.
(a) State the domain and range for $g(x)$. [2 marks]
(b) Show that $g(x)$ is a one-to-one function. Hence find $g^{-1}(x)$. [5 marks]
(c) On the same axis, sketch the graph of $g(x)$ and $g^{-1}(x)$. [3 marks]
5. (a) A function g is defined as $g(x) = \begin{cases} kx^2 - 12, & x \in [-2, 2], \\ \sqrt{x^2 - 4}, & x \notin [-2, 2]. \end{cases}$
Given that g is continuous for all values of x , find the value of k . [3 marks]
(b) A function h is defined as $h(x) = \frac{x+2}{\sqrt{x^2-4}}$. Find all the asymptotes of h . [6 marks]

6. (a) Given $y = e^{-x} \cos 2x$, show that $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$. [4 marks]

(b) Given $x = \frac{t}{t+1}$, $y = t - 1$, $t > 0$.

(i) Find $\frac{dy}{dx}$ in terms of t . Find the value of t if $\frac{dy}{dx} = 4$. [5 marks]

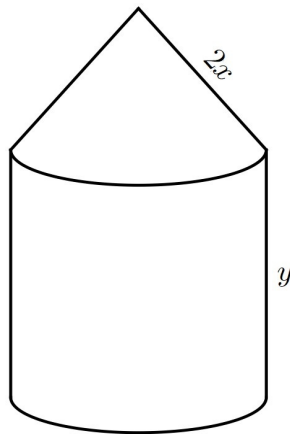
(ii) Find the tangent equation when $\frac{dy}{dx} = 4$. [4 marks]

7. An open cylindrical container can hold 16 cm^3 of rice. The radius of the container is $x \text{ m}$ and its height is $y \text{ m}$.

(a) Show that $y = \frac{16}{\pi x^2}$. [2 marks]

(b) Find the surface area of the container in terms of x . [2 marks]

(c) A conical cover of slant height $2x \text{ m}$ is placed on top of the container to form a silo as shown below.



Given that the cost per unit area of the cylindrical container is p ringgit per square metre. If the cost per unit area of the cover is 50% more than the cost per unit area of the cylindrical container, show that the cost, C , of the silo is $C = p \left(4\pi x^2 + \frac{32}{x} \right)$.

Hence, show that the total cost, C , is minimum when $x = \sqrt[3]{\frac{4}{\pi}}$. [9 marks]

END OF QUESTION PAPER

Answers**SECTION A**

1. (a) $\frac{5}{2}$ (b) $-\frac{1}{a^2}$

2. *DIY*3. Increasing intervals: $(0, 2)$ and $(2, \infty)$, Decreasing intervals: $(-\infty, -2)$ and $(-2, 0)$ **SECTION B**

1. $z = -5 + 2i$

2. (a) $x = 5$ (b) $x \in [-9, -1]$

3. (a) $f \circ f = \frac{4x}{x+1}$, $f \circ f$ is undefined when $x = -1$

(b) $f^{-1}(x) = \frac{-x}{2-x}$, $D_{f^{-1}} = (-\infty, 2) \cup (2, \infty)$

(c) $g(x) = x^2 + 2x$

4. (a) $D_f = (-\infty, \infty)$, $R_f = (1, \infty)$ (b) *DIY*, $g^{-1}(x) = \frac{\ln(x-1)}{\ln 2} + 2$ (c) *DIY*

5. (a) $k = 3$

(b) Vertical asymptotes: $x = 2$, $x = -2$ (b) Horizontal asymptotes: $y = 1$, $y = -1$

6. (a) *DIY* (b) (i) $t = 1$ (ii) $y = 4x - 2$

7. (b) $S = \pi x^2 + \frac{32}{x}$